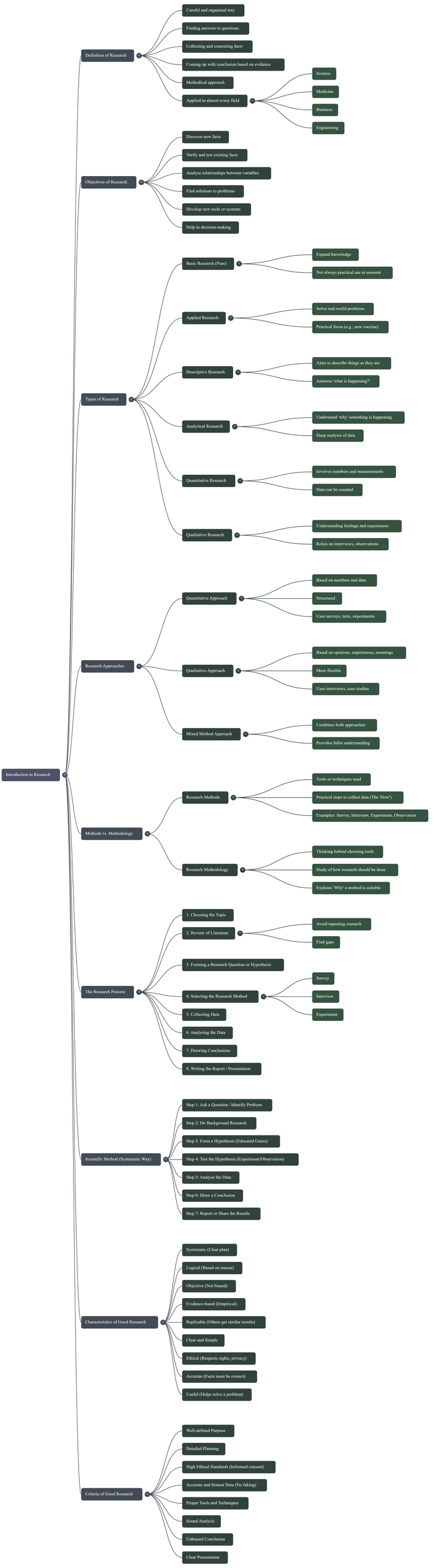
Introduction to Research
Definition of Research
Careful and organised way
Finding answers to questions
Collecting and examining facts
Coming up with conclusion based on evidence
Methodical approach
Applied in almost every field
Science
Medicine
Business
Engineering
Objectives of Research
Discover new facts
Verify and test existing facts
Analyse relationships between variables
Find solutions to problems
Develop new tools or systems
Help in decision-making
Types of Research
Basic Research (Pure)
Expand knowledge
Not always practical use at moment
Applied Research
Solve real-world problems
Practical focus (e.g., new vaccine)
Descriptive Research
Aims to describe things as they are
Answers 'what is happening?'
Analytical Research
Understand 'why' something is happening
Deep analysis of data
Quantitative Research
Involves numbers and measurements
Data can be counted
Qualitative Research
Understanding feelings and experiences
Relies on interviews, observations
Research Approaches
Quantitative Approach
Based on numbers and data
Structured
Uses surveys, tests, experiments
Qualitative Approach
Based on opinions, experiences, meanings
More flexible
Uses interviews, case studies
Mixed Method Approach
Combines both approaches
Provides fuller understanding
Methods vs. Methodology
Research Methods
Tools or techniques used
Practical steps to collect data (The 'How')
Examples: Survey, Interview, Experiment, Observation
Research Methodology
Thinking behind choosing tools
Study of how research should be done
Explains 'Why' a method is suitable
The Research Process
1. Choosing the Topic
2. Review of Literature
Avoid repeating research
Find gaps
3. Forming a Research Question or Hypothesis
4. Selecting the Research Method
Survey
Interview
Experiment
5. Collecting Data
6. Analysing the Data
7. Drawing Conclusions
8. Writing the Report / Presentation
Scientific Method (Systematic Way)
Step 1: Ask a Question / Identify Problem
Step 2: Do Background Research
Step 3: Form a Hypothesis (Educated Guess)
Step 4: Test the Hypothesis (Experiment/Observation)
Step 5: Analyse the Data
Step 6: Draw a Conclusion
Step 7: Report or Share the Results
Characteristics of Good Research
Systematic (Clear plan)
Logical (Based on reason)
Objective (Not biased)
Evidence-based (Empirical)
Replicable (Others get similar results)
Clear and Simple
Ethical (Respects rights, privacy)
Accurate (Facts must be correct)
Useful (Helps solve a problem)
Criteria of Good Research
Well-defined Purpose
Detailed Planning
High Ethical Standards (Informed consent)
Accurate and Honest Data (No faking)
Proper Tools and Techniques
Sound Analysis
Unbiased Conclusion
Clear Presentation